



Master in Quantitative Economics and Finance

**Generating Beliefs for Private Equity
using Large Language Models**

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Chapter 1

Introduction

1.1 Motivation

Private equity (PE) has experienced remarkable growth over the past two decades, emerging as a dominant force in institutional portfolios. Global assets under management in PE exceeded USD 17 trillion in 2024 (see 1.1b), with pension funds, sovereign wealth funds, and endowments allocating an increasing share of capital to the asset class. This expansion is driven by the promise of superior returns and diversification benefits; however, PE remains fundamentally different from public markets. Valuations are opaque, transactions are infrequent, and information is primarily controlled by general partners (GPs) through periodic reporting. These features make it difficult to determine whether observed performance reflects true skill or risk-taking, and they place investor beliefs at the center of capital allocation decisions.

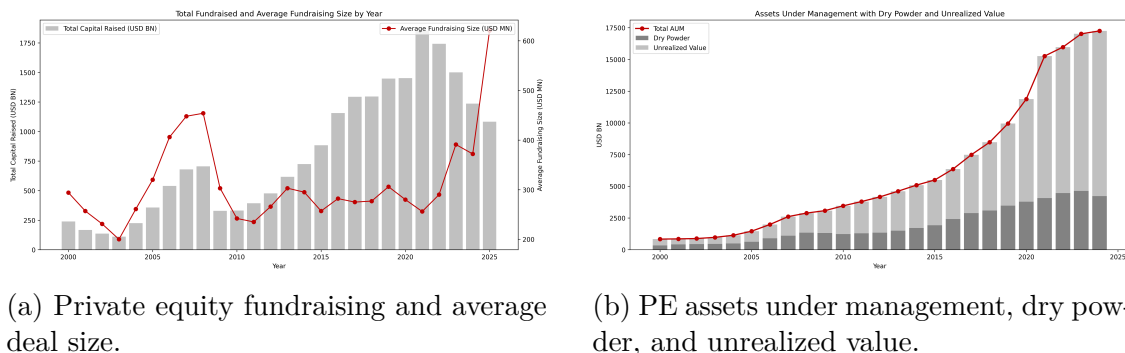


Figure 1.1: Trends in private equity capital and dry powder levels over time. Data from Preqin.

Unlike public equities, where investor beliefs can be inferred directly from market prices, private equity beliefs are not observable in real time. Limited partners (LPs) form expectations about future distributions, exit values, and returns by interpreting a narrow set of signals: GP reports, due diligence materials, industry news, and historical cash flows. These communications often combine hard information with softer language that can influence perceptions. It remains unclear whether LPs are consistently able to extract the predictive content of these narratives or if they are sometimes influenced by sentiment and bias.

Historical episodes illustrate the stakes. During the global financial crisis (2008–2009), many PE funds were slow to revise net asset values (NAVs), creating a disconnect between reported valuations and underlying fundamentals. In contrast, during the COVID-19 shock, GPs initially issued cautious updates but quickly turned more optimistic as public markets recovered. Both episodes suggest that GP communications can shape investor beliefs in ways not fully explained by fundamentals, with potential implications for capital allocation and risk perception (see Figure 1.2).

Recent advances in natural language processing (NLP) and the emergence of large language models (LLMs) offer a new lens to examine these dynamics. LLMs can process complex textual information and generate probabilistic forecasts based on contextual cues, mimicking or even surpassing human interpretation. In public markets, LLMs have already been shown to match human analysts performances

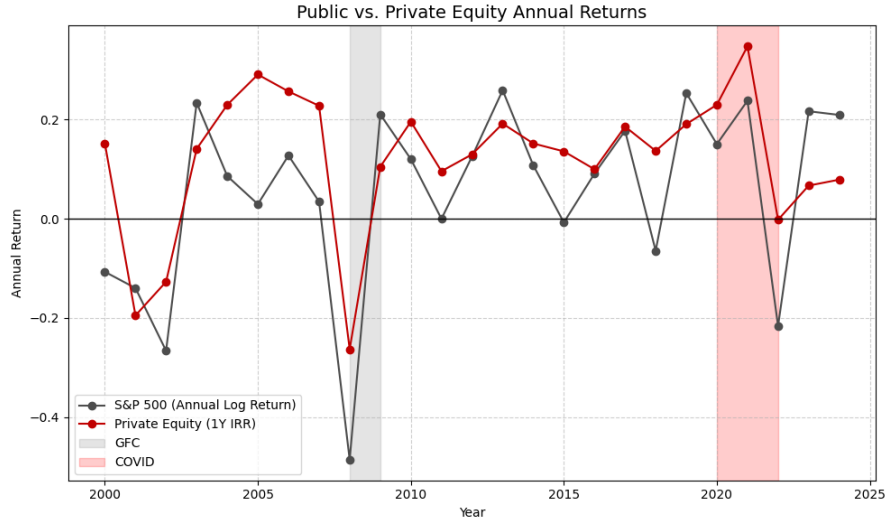


Figure 1.2: Private equity returns (source: Preqin) are calculated using the NAV reported by GPs. Therefore, they tend to be much less volatile than market returns.

when it comes to predict returns, infer sentiment, and identify risk exposures from unstructured data. However, private markets present a more challenging environment: the data is scarcer, disclosures are less standardized, and outcomes are realized over long horizons. The aim of this thesis is to introduce a methodology to use LLMs to process GPs report and extract information effectively and quantify the sentiment expressed in these documents.

1.2 Literature Review

The starting point for this thesis is the literature on belief formation in asset pricing. Classical models assume rational expectations, where investors form unbiased forecasts given available information. Yet empirical evidence shows that beliefs often deviate systematically from fundamentals. Behavioral models document patterns such as overreaction to recent performance, underreaction to fundamentals, and herding behavior (Barberis, 2018), observed in analyst forecasts, mutual fund flows, and investor reactions to earnings announcements. In public markets, these biases are frequently linked to the use of heuristics when processing complex signals.

Bybee (2025) introduces a framework in which belief formation is shaped by language patterns. Comparing human analyst reports and LLM-generated forecasts in response to news, he finds that both humans and models overweight recent returns, anchor on past price movements, and underweight fundamentals. This suggests that beliefs can be modeled as processes influenced by the statistical patterns embedded in language rather than purely rational inference.

Private equity adds further complexity. Unlike public equities, PE valuations are not continuously updated, and information flows are limited to periodic GP reports and occasional external assessments. Recent work emphasizes how investor beliefs and biases shape asset prices, even in settings where fundamentals are difficult to observe. For instance, ? highlight the role of subjective beliefs in private equity valuation, showing that reported performance can diverge from underlying fundamentals depending on how narratives are interpreted. Similarly, ? study how institutional investors select asset managers, pointing to the importance of qualitative signals in decision-making. Reported NAVs may be smoothed or manipulated, particularly around fundraising events or during market downturns (Brown et al., 2016). Gupta and Van Nieuwerburgh (2020) show that when PE cash flows are decomposed into exposures to risk factors, most apparent outperformance disappears, and risk-adjusted returns are often negative. This indicates that beliefs based solely on NAVs are likely distorted. The scarcity and opacity of information in private markets amplify the role of narratives and subjective judgment, making them a natural extension of Bybee’s framework.

1.2.1 LLMs in Finance

The rapid development of large language models has opened new avenues for extracting predictive signals from unstructured financial data. Early applications have focused on public markets, where extensive news, analyst reports, and earnings calls provide rich training grounds. Studies show that LLMs can generate forecasts

that incorporate complex qualitative information, often matching or exceeding traditional models in sentiment analysis and event prediction (??). However, these applications rely on abundant and standardized textual data—conditions that do not hold in private markets.

Fernández Tamayo et al. (2023) apply machine learning to textual disclosures in private placement memorandums. They find that while investors often fail to use qualitative signals effectively, algorithms leveraging textual features achieve strong predictive power for fund performance. Nuanced textual information, such as references to operational expertise, networks, or strategy, contains latent signals of skill that human investors often overlook. This underscores the potential of AI-driven analysis in opaque markets and motivates the application of LLMs to belief formation in private equity.

1.3 Research Question and Contributions

Against this backdrop, the central research question of this thesis is:

Are LLMs able to understand PE investment reports and form quantitative beliefs? How accurate are they?

To address this question, I leverage a dataset of over 20,000 PE investment reports from Preqin, enriched with cash flow data on capital calls and distributions. The reports provide information on portfolio companies, market outlooks, and fund strategies in the unstructured format that LLMs are designed to process. By extracting belief measures directly from this text and comparing them to realized outcomes, I can test whether PE disclosures contain predictive content and whether LLM-generated beliefs exhibit systematic biases.

This thesis makes three contributions:

1. **Measurement of Beliefs:** I extend Bybee’s methodology to quantify beliefs

in private equity by applying LLMs to GP communications and extracting structured forecasts from unstructured text.

2. **Implications for Asset Pricing and Allocation:** By linking text-derived beliefs to realized cash flows and factor-adjusted performance, I provide evidence on whether beliefs contain incremental information for forecasting PE outcomes.
3. **Grounds for further research:** I provide ideas for further research.

Chapter 2

Data & Setup Description

2.1 Selected fund managers

I consider a list of 20 GPs and 20 LPs, selected by fund size and reported in table 2.1.

For each of them, all the investor documents that contain the fund name are obtained from Preqin, and stored for LLM processing. Finally, for the 20 GPs, the cash flows of the years 2014–2024 for the funds they manage are also obtained. The cash flows database contains, complemented by fund metadata

2.2 LLM and Computational Setup

When working with large language models (LLMs), the choice of hardware is crucial, as even modestly sized models require substantial GPU resources. Insufficient computational power can limit access to state-of-the-art models and reduce experimental flexibility. For this analysis, I employ DeepSeek-r1, an open-weights, reasoning-focused, LLM introduced by the DeepSeek AI research team in late 2024 and officially released in January 2025. The model quickly became a disruptive force in the AI landscape, even contributing to a significant selloff in Nvidia’s stock by demonstrating strong performance on relatively modest hardware. In particular, I rely on the compact version, *DeepSeek-r1:8B* (8 billion parameters), which balances

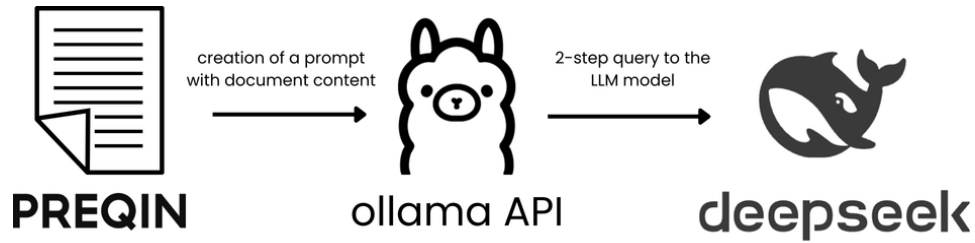


Figure 2.1: Workflow schematic with main components.

computational efficiency with high reasoning capabilities (see 2.2).

Model			
architecture	qwen3		
parameters	8.2B		
context length	131072		
embedding length	4096		
quantization	Q4_K_M		
Capabilities			
completion			
thinking			
Parameters			
stop	"<	beginofsentence	>"
stop	"<	endofsentence	>"
stop	"<	User	>"
stop	"<	Assistant	>"
temperature	0.6		
top_p	0.95		
License			
MIT License			
Copyright (c) 2023 DeepSeek			

The model is deployed locally on the hardware of choice, an MSI GS65 laptop mounting an NVIDIA GeForce GTX 1070 GPU with 8 GB of VRAM (CUDA 12.8), through Ollama, a lightweight framework that lets you run large language models locally on your machine with simple commands and efficient resource management. Particularly, *Ollama* exposes an API through which the locally installed model can be queried by sending requests containing a prompt expressed in natural language. For this analysis, the prompt consists in textual content from the investor documents, and instructions on how to process it (depicted in Fig. ??).

Table 2.1: Top 20 Private Equity GPs and LPs by Fund Size or Commitment (USD mn)

GP	Fund Size	LP	Commitment
TPG	57,088.5	California Public Employees' Retirement System	29,685.33
The Carlyle Group	53,275.79	CPP Investments	25,130.54
Apollo Global Management	52,034	California State Teachers' Retirement System	22,637.9
Blackstone	46,950	Washington State Investment Board	16,853.35
CVC Capital Partners	38,355.47	Oregon Public Employees Retirement Fund	16,093.9
Apax Partners	35,544.61	New York State Common Retirement Fund	15,128.15
Goldman Sachs Asset Management	30,550	Teacher Retirement System of Texas	14,248.47
Bain Capital	29,507	Florida State Board of Administration	11,212.6
Kohlberg Kravis Roberts	28,572.18	Hamilton Lane	9,833.73
Advent International	27,559.44	Pennsylvania Public School Employees' Retirement System	8,414.04
Permira	25,125.9	Michigan Department of Treasury	8,283.78
Silver Lake	23,300	State of Wisconsin Investment Board	6,602.43
Providence Equity Partners	22,808	Massachusetts Pension Reserves Investment Management Board	6,277.31
Madison Dearborn Partners	22,163	Teachers Retirement System of Georgia	5,321.97
EQT	19,486.98	Teachers' Retirement System of the State of Illinois	5,215.23
Nordic Capital	16,014.86	State Teachers Retirement System of Ohio	5,204.57
Welsh, Carson, Anderson & Stowe	15,386.7	Los Angeles County Employees' Retirement Association	5,058.98
Leonard Green & Partners	15,007	New York State Teachers' Retirement System	4,736.66
Charterhouse Capital Partners	14,468.42	Maryland State Retirement and Pension System	4,675
Onex	13,810	Minnesota State Board of Investment	4,596.94

Table 2.2: Number of retrieved documents

LP	Number of documents
California Public Employees' Retirement System	359
CPP Investments	103
California State Teachers' Retirement System	142
Washington State Investment Board	84
Oregon Public Employees Retirement Fund	114
New York State Common Retirement Fund	139
Teacher Retirement System of Texas	127
Florida State Board of Administration	247
Hamilton Lane	1571
Pennsylvania Public School Employees' Retirement System	71
Michigan Department of Treasury	73
State of Wisconsin Investment Board	147
Massachusetts Pension Reserves Investment Management Board	56
Teachers Retirement System of Georgia	53
Teachers' Retirement System of the State of Illinois	246
State Teachers Retirement System of Ohio	137
Los Angeles County Employees' Retirement Association	209
New York State Teachers' Retirement System	84
Maryland State Retirement and Pension System	122
Minnesota State Board of Investment	96
Total	4180

GP	Number of documents
TPG	810
The Carlyle Group	455
Apollo Global Management	687
Blackstone	1403
CVC Capital Partners	1014
Apax Partners	800
Goldman Sachs Asset Management	947
Bain Capital	1241
Kohlberg Kravis Roberts	485
Advent International	757
Permira	903
Silver Lake	908
Providence Equity Partners	283
Madison Dearborn Partners	89
EQT	2719
Nordic Capital	388
Welsh, Carson, Anderson & Stowe	617
Leonard Green & Partners	425
Charterhouse Capital Partners	140
Onex	895
Total	15966

Chapter 3

Methodology

This chapter presents the methodology used in this research.

3.1 2-query procedure to generate beliefs

The scope of this procedure is to produce quantitative estimates from textual content. In order to do so, we need to ask the LLM to produce structured results (which can be seen as *answers to a survey*), which are then aggregated to make the index. Due to hardware limitations, the whole context length (the amount of input the model can see in one run, or the *short term memory* of an LLM) of 131072 tokens (the chunks in which the text is split, each roughly corresponding to $\frac{3}{4}$ of an average English word) that Deepseek-r1 would support is not available. Instead, Ollama restricts the model to a 2048 token window, and batches the requests. Therefore, the model is unable to handle complex tasks, such as providing a well-structured output, in just one go (the instructions *get lost* in the sea of information of the document text). Instead, first each document is summarized to few lines with a particular focus on the quantities of interest, and then a second round produces the structured output.

Document summary: All the documents in table 2.1 are summarized focusing on the most important pieces of information. An example of the prompt used for GPs is reported in 3.1. Examples of output can be found in section ??

```
You are a financial analyst.
Your task: summarize the following investment report,
keep track of the name of the fund and the date of the report,
focus on {quantity_of_interest} for the {period} in the perspective of fund
manager {fund_manager}, its magnitude in percentage terms, and your
confidence in the prediction.

Context from the document:
\"\"\"{text}\"\"\"

remember your task: summarize the following investment report,
keep track of the name of the fund and the date of the report,
focus on {quantity_of_interest} for the {period} in the perspective of fund
manager {fund_manager}, its magnitude in percentage terms, and your
confidence in the prediction.
```

JSON-formatted answers: Then, I execute a second round of queries, this time asking for the forecast. A prompt example is the following:

```
You are a financial analyst that must output only a valid JSON object,
forecasting {variable_of_interest} for the {period}.

Context from the document:
\"\"\"{summary_text}\"\"\"

Based on the above context, produce strictly one JSON object following
this schema:
{json_schema}

Respond with ONLY the JSON object.
```

The machine is also told that the answer must be submitted in a JSON format as follows:

```
{"results": {
  "fund_name": "<fund name>",
  "report_date": "<yyyymmdd>",
  "outlook": "<increase|stable|decrease>",
  "magnitude": <percentage>,
  "confidence": <number between 0 and 1>,
  "explanation": "<short one-line reasoning>"
}
```


3.2 Construction of the Sentiment Index

Finally the entries from the JSON files are extracted and collected in a database. Moreover, the outlook is converted to a numerical value according to the following table:

increase	1
stable	0
decrease	-1

Following Bybee’s Ghost in the Machine Approach, the numerical outlooks O_t^m for each GP m are then averaged based on the report date to create the sentiment index S_t^m . I use both yearly and quarterly aggregation.

$$S_t^m = \sum_{t < u \leq t+1} O_u^m, \quad (3.1)$$

where t is the period. The other values extracted from the beliefs are not used.

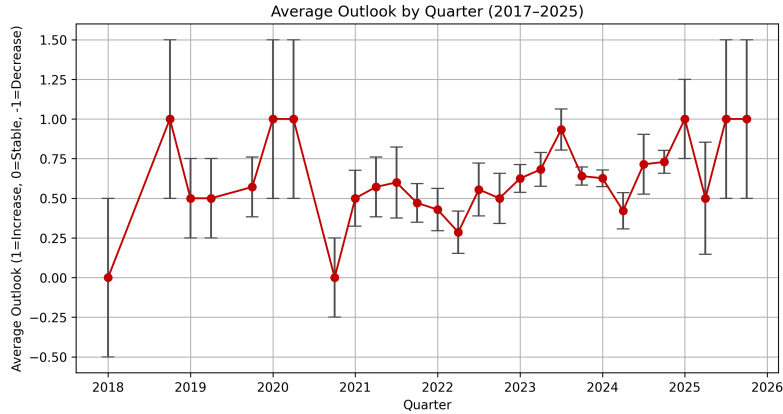


Figure 3.1: The sentiment index for GP *Carlyle Group*

3.3 Comparison with hard data

The aim is to be able to run a panel data regression to evaluate how well does the predicted sentiment correlates with the actual data for the same quantity (which

in this analysis, are outlooks on cash flow distributions), as well as other macroeconomic indicator, such as the aggregated returns of public and private markets.

Net cashflows The data for net cashflows, i.e. sum of inflows and distributions, net of fees, is directly available on Preqin. The values are discounted to the beginning of each period and aggregated to the desired frequency to match the sentiment 3.1. I tested both the levels and the changes, the latter both in relative and absolute terms

Market returns As a proxy for public and private market returns I use the S&P500 log-returns and Preqin aggregate IRR, respectively, as in Fig. 1.2.

Chapter 4

Results

4.1 Case Study

In this section I detail the intermediate outputs of the index construction for the GP *TPG*. For this GP, 810 documents have been processed, including the document I will pick as an example, 1047 - Oregon Public Employees Retirement Fund Opportunity Portfolio, which details the performance of several funds in which the Oregon Public Employees Retirement Fund is invested, managed by TPG. The first step of the elaboration is the summary, which I present below as produced by the model:

```
### Summary of Investment Report (Focusing on TPG's Perspective)

The report provides an overview of several private equity partnerships
managed by TPG, including the TPG Opportunity Fund (I), TPG
Specialty Lending Fund, and TPG Specialties LP. Below is a summary
of each fund's performance as of December 31, 2020:

#### **TPG Opportunity Fund (I)**
- **Total Value Multiple**: 1.43x
- **Time-Weighted Return (TWR)**: 6.1%
- **Performance Perspective**: The fund is performing well with a
  strong total value multiple and modest growth rate.

#### **TPG Specialty Lending Fund**
- **Total Value Multiple**: 2.75x
- **Time-Weighted Return (TWR)**: 15.0%
- **Performance Perspective**: The fund shows significant upside
  potential, driven by favorable performance in its loan portfolio.

#### **TPG Specialties LP**
- **Total Value Multiple**: 1.38x
- **Time-Weighted Return (TWR)**: 8.0%
- **Performance Perspective**: The fund has modest growth but is
  lagging compared to the Specialty Lending Funds due to lower-than-
  expected performance in some of its loan portfolio.
```

Manager's Perspective on Capital Distributions

TPG's manager believes that capital distributions for the year following the report date (2021) will vary across funds based on their current performance and structural characteristics:

1. **TPG Opportunity Fund (I)**: The fund is expected to receive moderate capital distributions, commensurate with its strong performance. TPG anticipates that the fund's capital distribution timing aligns closely with its performance momentum.
2. **TPG Specialty Lending Funds**: These funds are projected to experience higher capital distributions due to their significant upside potential and favorable loan portfolio performance. TPG expects these funds to maintain their leading position in terms of capital distribution magnitude.
3. **TPG Specialties LP**: The fund is expected to receive lower capital distributions compared to the Specialty Lending Funds, reflecting its modest growth trajectory.

Confidence Level

- **For TPG Opportunity Fund (I)**: High confidence due to strong performance and stability.
- **For TPG Specialty Lending Funds**: Very high confidence given their favorable loan portfolio performance.
- **For TPG Specialties LP**: Moderate confidence due to lower upside potential compared to the Specialty Lending Funds.

Conclusion

TPG anticipates that capital distributions will be distributed in a manner that reflects each fund's current performance. The manager is optimistic about the opportunities for growth across all funds but acknowledges that diversification among these funds could lead to varying distribution timings and magnitudes.

This summary provides TPG's perspective on its investment strategy, emphasizing the importance of managing multiple private equity partnerships with different performance characteristics to achieve steady capital distribution outcomes in 2021.

The output of the first phase is then fed once again to the model, to produce the following JSON output:

```
{
  "results": [
    {
      "fund_name": "TPG Opportunity Fund (I)",
      "report_date": "2021",
      "outlook": "increase",
      "magnitude": 5,
      "confidence": 0.8,
      "explanation": "The fund is expected to receive moderate capital distributions due to its strong performance momentum."
    }
  ]
}
```

```

    },
    {
      "fund_name": "TPG Specialty Lending Fund",
      "report_date": "2021",
      "outlook": "increase",
      "magnitude": 10,
      "confidence": 0.95,
      "explanation": "The fund is projected to experience
                     higher capital distributions due to its significant
                     upside potential."
    },
    {
      "fund_name": "TPG Specialties LP",
      "report_date": "2021",
      "outlook": "increase",
      "magnitude": 4,
      "confidence": 0.6,
      "explanation": "The fund is expected to receive lower
                     capital distributions due to its modest growth
                     trajectory."
    }
  ]
}

```

Then, all the entries from the JSON files are compiled in a database, for a total of 880 entries, as most documents focus on a single fund and some do not contain relevant information. After aggregation, index shown in Fig. 4.1 is produced:

Finally, after filtering points with too few observations, an OLS regression is run on the actual data that the index is meant to represent. The results of the regression using yearly aggregation, reported in Table 4.1, show a considerable and significant collinearity between the index and the actual data.

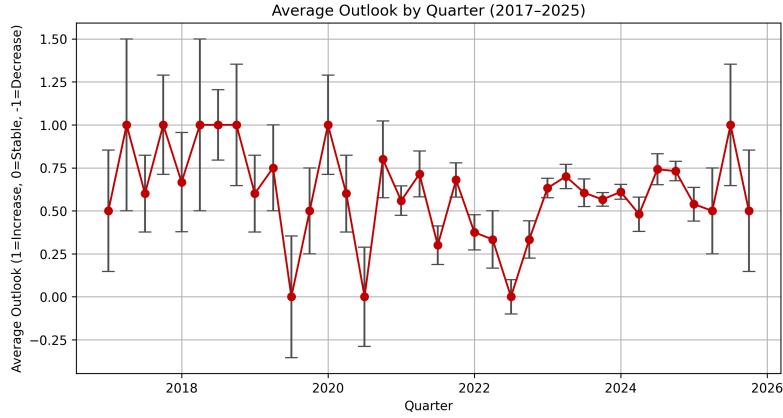


Figure 4.1: Quarterly aggregation for the sentiment index for the *TPG* GP. Error bar sizes are inversely proportional to the number of available data entries for each quarterly point.

Table 4.1: OLS Regression of Sentiment on Net Cashflow Changes

	Coef.	Std. Err.	t	P> t
const	0.6471	0.037	17.614	0.000
rel_change_net_cf	0.5885	0.117	5.038	0.015
<i>Observations:</i> 5, R^2 : 0.894, <i>Adj. R</i> ² : 0.859				
<i>F</i> -statistic: 25.38 (p = 0.015), <i>DW</i> : 1.873				

4.2 Panel Regression

4.3 Comparison with Market Baselines

A primary concern is the potential presence of look-ahead bias, meaning the language model may have been trained on the same information later used to construct the sentiment index. While it is highly unlikely that the model was trained on proprietary Preqin documents, its forecasts could still be influenced by publicly available information covering similar content. To address this concern, an initial test was conducted by regressing the sentiment index on macroeconomic variables—factors that are almost certainly part of the model’s training data. The results showed no significant correlation (see Table 4.2), suggesting that the index does not simply reflect pre-existing knowledge of macro trends.

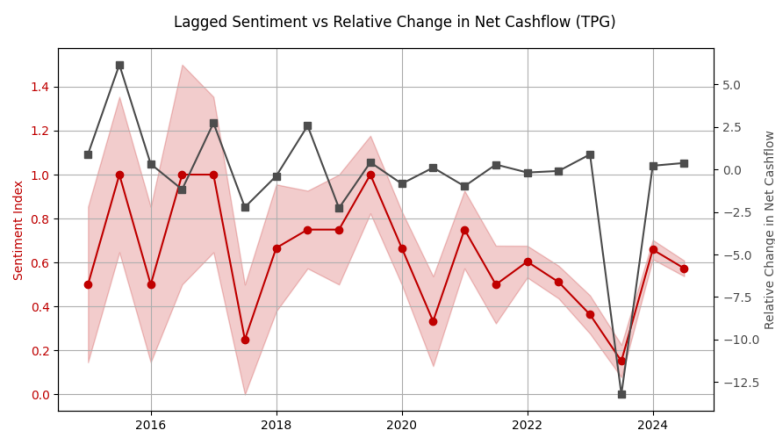


Figure 4.2: Comparison between the sentiment index and the data from Preqin

Table 4.2: OLS Regression Results

Variable	Coefficient	Std. Error	t-Stat (p-value)
const	0.7880	0.112	7.005 (0.000)
snp500	0.4859	0.495	0.981 (0.348)
irr	-1.2399	0.820	-1.513 (0.159)

Model Statistics:

R-squared: 0.173 Adj. R-squared: 0.023

F-statistic: 1.153 Prob(F-statistic): 0.351

Observations: 14 Durbin-Watson: 1.398

Chapter 5

Conclusion and Future Work

5.1 Key Takeaways

The proposed methodology demonstrates strong potential when the sentiment index is based on a sufficient number of observations, ensuring that the resulting measure is both stable and representative. This analysis suggests that in private equity, certain general partners (GPs) appear more consistent in maintaining the expectations they set, while others exhibit greater variability, potentially reflecting differences in communication strategies or internal forecasting accuracy. However, one critical limitation is computational efficiency: processing large volumes of unstructured text data can become prohibitively time-consuming without access to adequate computing resources, especially when applying advanced language models. Another challenge lies in the heterogeneity of the dataset. Mixing funds across asset classes and incorporating statements from both GPs and limited partners (LPs) can introduce noise, making it harder to draw clear causal inferences. This highlights the importance of segmenting the analysis by fund type, region, or strategy to improve robustness and interpretability. Despite these limitations, the methodology is highly promising. It is inherently scalable and can be adapted to a wide range of applications, from monitoring market sentiment in real time to enhancing risk

models for capital allocation. As computational infrastructure improves and more granular data becomes available, this approach could become a standard tool for analyzing belief formation and expectation management in private markets.

5.2 Roadmap for Extension

Here I present some extensions to overcome the limitations found in my research.

Deployment on ULHPC and usage of more advanced models On the local machine, inference was carried out on an NVIDIA GeForce GTX 1070 GPU with 8 GB of VRAM (CUDA 12.8), which limited the feasible model size and imposed a maximum context length of 2,048 tokens for Deepseek 8B. This required documents to be split into multiple segments for processing. In contrast, the ULHPC nodes are equipped with NVIDIA Tesla V100-SXM2 GPUs with 32 GB of VRAM (CUDA 12.7), offering four times the memory capacity and significantly greater computational capacity. This would enable the use of both Deepseek 8B and larger model such as GPT-OSS 20B with extended context lengths of up to 131,072 tokens, allowing entire documents to be processed in a single pass, thereby reducing pre-processing overhead and improving throughput. In particular, it would significantly reduce the computational time needed.

Model	
architecture	gptoss
parameters	20.9B
context length	131072
embedding length	2880
quantization	MXFP4
Capabilities	
completion	
tools	
thinking	
Parameters	
temperature	1
License	
Apache License	
Version 2.0, January 2004	

Investigate other quantities and other forecasting horizons This research was limited to forecasting changes in distribution cashflows in the next year. With better hardware and more time to experiment, it will be possible to ask about other quantities, such as returns and fundraising, and different time horizons, such as the next quarter, or the next 2 years.

Improve Granularity While the aggregation per GP provides an immediate overview of the performance of its funds, in reality it often aggregates funds from different asset classes (such as Buyout, Venture Capital, Real Estate and Fund of Funds) which often perform in very different ways. Moreover, there are other important factors that influence the performances of a fund, for instance the Vintage effect, i.e. the years since its inception which determine at which stage of its life the fund is. Therefore, it would be more appropriate to create indices for each fund and then aggregate appropriately per GP or Asset Class.

Appendix A

Appendices

A.1 Additional Tables and Figures

In this section I include additional figures.

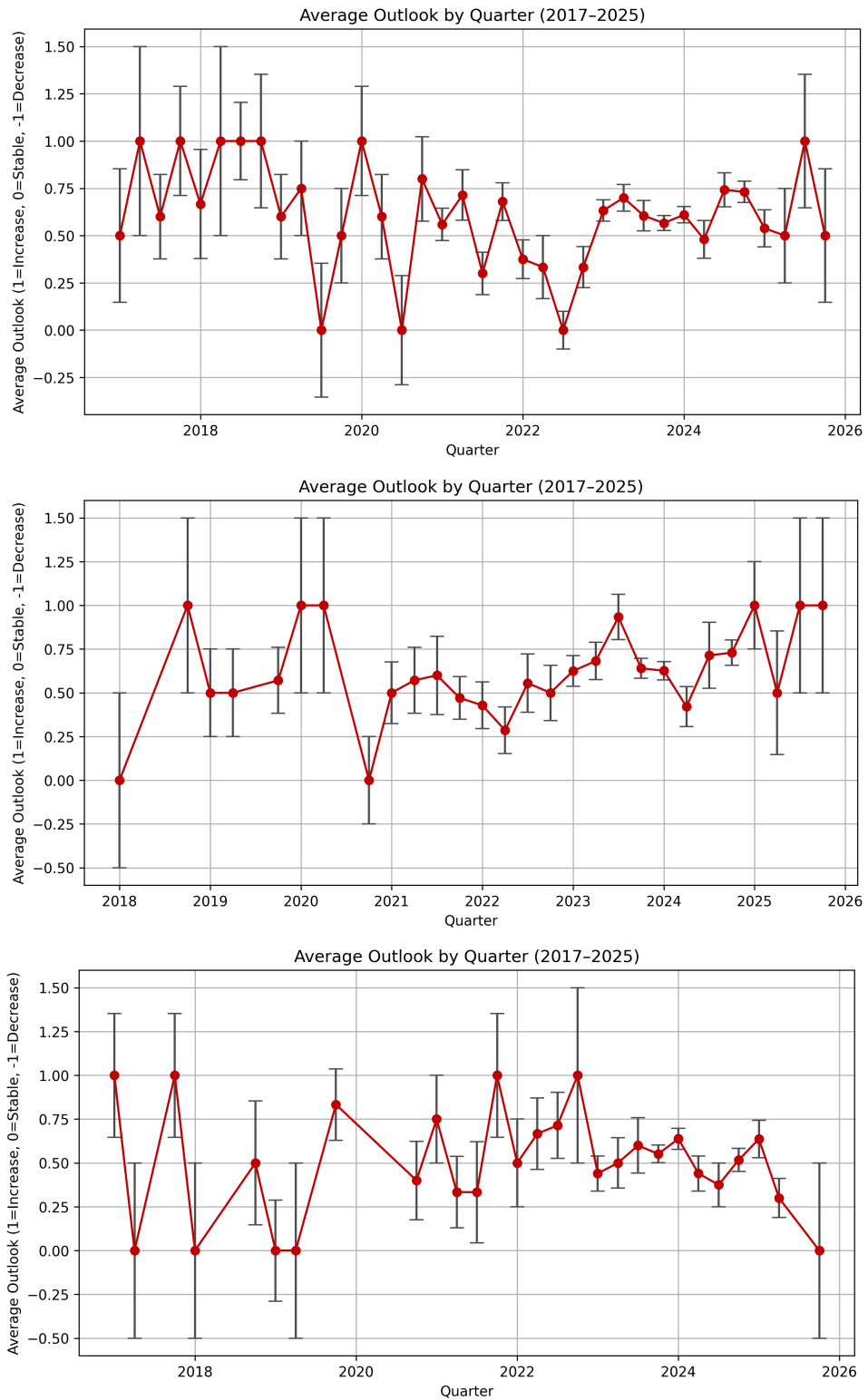


Figure A.1: The sentiment indices for TPG, Carlyle Group and Apollo Global Management. Quarterly aggregation.

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